
IK 548: Advanced NLP Techniques for Indian Languages

Course Code	IK 548
Course Name	Advanced NLP Techniques for Indian Languages
L-T-P-C	3-0-0-3
Intended for	BTech/MTech/MS/MSc/MA/Ph.D
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Natural Language Processing with Classification and Vector Spaces**
 - Sentiment Analysis with logistic Regression (Total = 6 Hours)
 - * 15 videos - Total 84 minutes
 - * 13 readings - Total 100 minutes ● 1 quiz - Total 30 minutes
 - Sentiment Analysis with Naive Bayes (6 Hours)
 - * 13 videos - Total 44 minutes
 - * 12 readings - Total 111 minutes
 - * 1 quiz - Total 30 minutes
 - Vector Space Models (5 hours)
 - * 10 videos - Total 28 minutes
 - * 10 readings - Total 91 minutes
 - * 1 quiz - Total 30 minutes
 - Machine Translation And Document Search (6 Hours)
 - * 11 videos - Total 68 minutes
 - * 10 readings - Total 91 minutes
 - * 1 quiz - Total 30 minutes
- **Natural Language Processing with Probabilistic Models**
 - Autocorrect (4 hours)
 - * 11 videos - Total 31 minutes
 - * 10 readings - Total 37 minutes
 - * 1 quiz - Total 30 minutes
 - Part Of Speech Tagging and Hidden Markov Models (5 hours)
 - * 13 videos - Total 42 minutes
 - * 12 readings - Total 66 minutes
 - * 1 quiz - Total 30 minutes
 - Autocomplete and Language Models (5 hours)
 - * 11 videos - Total 53 minutes
 - * 10 readings - Total 70 minutes
 - * 1 quiz - Total 30 minutes
 - Word embedding with neural networks (8 hours)
 - * 22 videos - Total 73 minutes
 - * 22 readings - Total 88 minutes
 - * 1 quiz - Total 30 minutes
- **Natural Language Processing with Sequence Models**
 - Neural Networks For Sentiment Analysis (6 hours)

- * 11 videos - Total 39 minutes
- * 10 readings - Total 51 minutes
- * 1 quiz - Total 30 minutes
- Recurrent Neural Networks For Language Modeling (6 hours)
 - * 10 videos - Total 28 minutes
 - * 9 readings - Total 44 minutes
 - * 1 quiz - Total 30 minutes
- LSTMs and Named Entity Recognition (5 hours)
 - * 8 videos - Total 25 minutes
 - * 10 readings - Total 53 minutes
 - * 1 quiz - Total 30 minutes
- Siamese Networks (6 hours)
 - * 10 videos - Total 35 minutes
 - * 10 readings - Total 50 minutes
 - * 1 quiz - Total 30 minutes
- **Natural Language Processing with Attention Models**
 - Neural Machine Translation (7 hours)
 - * 15 videos - Total 88 minutes
 - * 6 readings - Total 37 minutes
 - * 1 quiz - Total 30 minutes
 - Text Summarization (5 hours)
 - * 10 videos - Total 39 minutes
 - * 8 readings - Total 62 minutes
 - * 1 quiz - Total 30 minutes
 - Question Answering (11 hours)
 - * 16 videos - Total 60 minutes
 - * 16 readings - Total 242 minutes
 - * 1 quiz - Total 30 minutes
 - Chatbot (6 hours)
 - * 9 videos - Total 63 minutes
 - * 13 readings - Total 141 minutes
 - * 1 quiz - Total 30 minutes

Text Books

1. Sue Knight, **NLP at Work**
2. Shelle Rose Charvet, **Words that change minds.**
3. Robert Dilts, **Sleight of mouth.**
4. Lynne Cooper, **Business NLP for dummies.**
5. **Mind line**

References

1. NA